

PROJECT ASSISTANT

An Adaptive Intelligence Framework for Data-Driven
Optimization of Collaborative Software Project
Environments



Team A



IT4010 PROGRESS PRESENTATION 2 (PP2) 25-26J-374

PROJECT ASSISTANT



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PROJECT ASSISTANT



PROBLEM STAKEHOLDERS

Context

Final-year and academic software projects require effective group formation, accurate supervisor matching, and timely progress evaluation. However, in many university environments, these activities are handled manually, making the process inefficient, subjective, and difficult to monitor.

STATEMENT

Stakeholders

Primary users – Students, Supervisors, Lecturers

Secondary users – Academic Departments, Universities,

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Observed issue

Manual group formation can create skill imbalance and weak collaboration. Supervisor assignment may not reflect domain expertise, and progress-related risks are often detected only after performance has already declined.

Why it matters

This affects fairness, transparency, project success, and timely intervention. It may result in mismatched teams, reduced guidance quality, missed deadlines, and poor academic outcomes.



RESEARCH GAP & OBJECTIVES

Gap

Existing solutions in academic project management do **not offer an integrated framework** that supports intelligent team formation, effective supervisor allocation, and continuous progress monitoring with predictive risk detection. As a result, **decision-making remains manual, inconsistent, and less effective.**

Objectives

- Optimizes student group formation using data-driven analysis
- Improves supervisor matching through semantic and domain-based alignment
- Monitors project progress using predictive analytics
- Detects risks and supports early intervention for better project outcomes



SOLUTION OVERVIEW

Solution

College students between 18–24 years old, balancing academics, part-time jobs, and personal life.

Key features

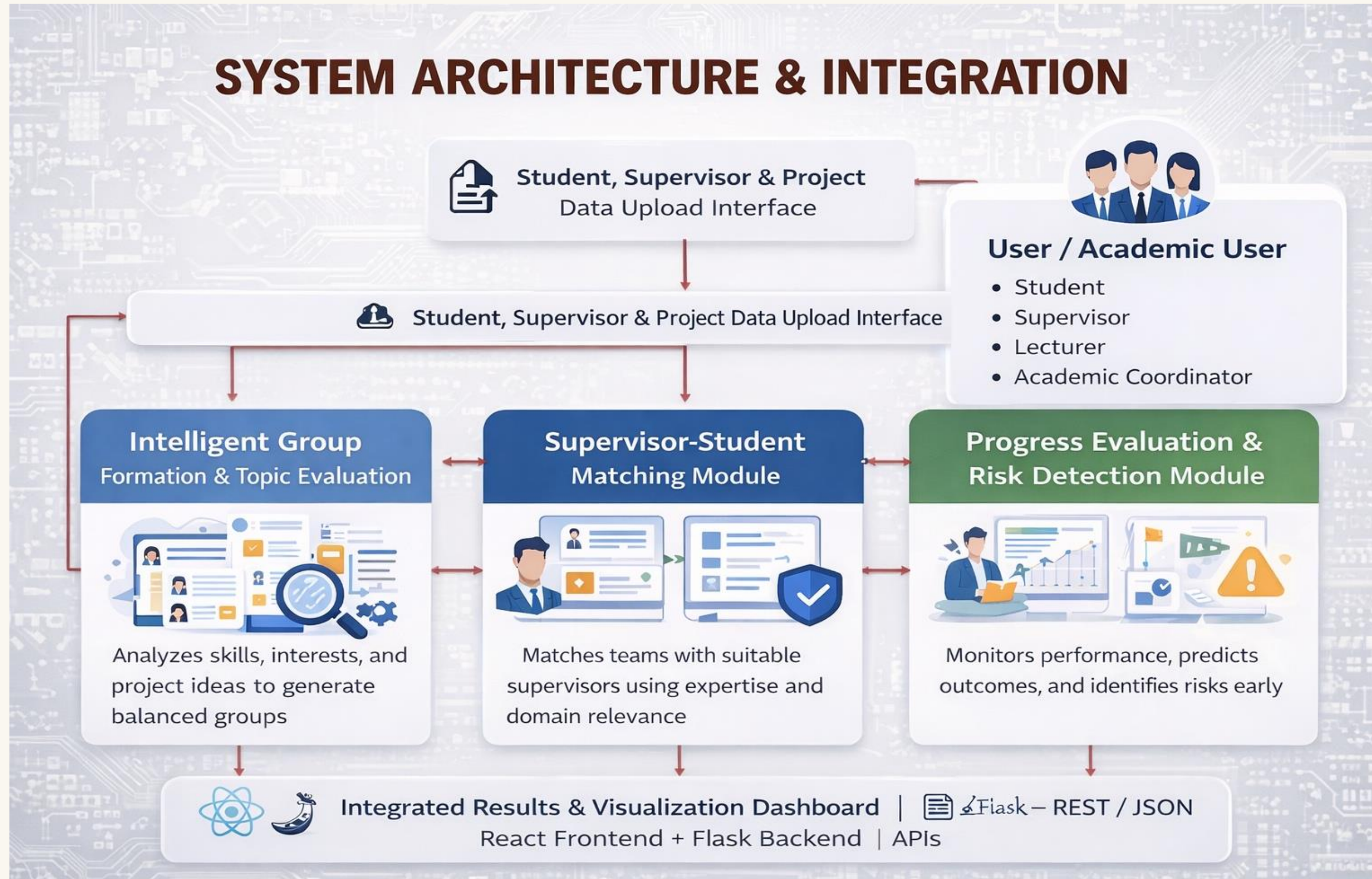
- intelligent group formation
- supervisor-student matching
- NLP Based project Guidance
- progress tracking + prediction
- risk detection + dashboards

Non-functional focus

- usability
- fairness
- transparency
- scalability



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COMMERCIALIZATION

Target users

Students, supervisors, lecturers, academic coordinators, and universities managing software or research-based projects.

Value proposition

Faster, fairer, and more intelligent academic project management through balanced group formation, better supervisor matching, and early risk detection.

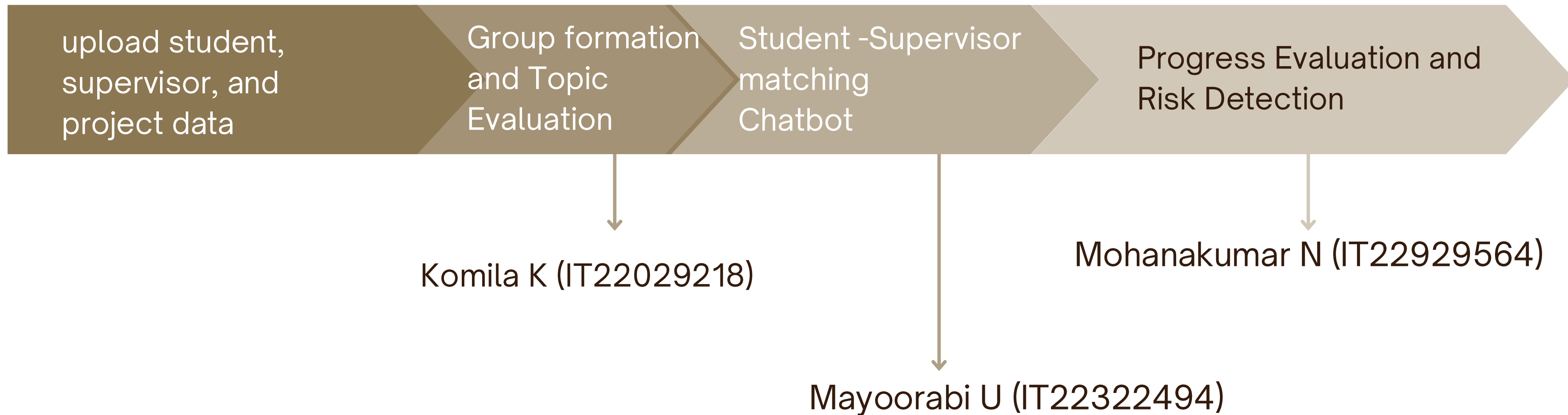
Revenue & Investment Recovery

Institutional subscriptions for universities and higher education institutes

Premium dashboard features for advanced analytics, reporting, and monitoring

Future integration support for LMS and academic management systems

DEMO PLAN



The End